

# A Packing of 63 Planes in $\mathbb{R}^{14}$

Limin Ge\*

October 8, 2026

## Abstract

We give 63 2-dimensional subspaces of  $\mathbb{R}^{14}$  whose smallest squared chordal distance is at least 1.728603. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 1.727385, obtained by deleting 1 of the 64 planes of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

## 1 Result

**Theorem.** *There are 63 subspaces of dimension 2 in  $\mathbb{R}^{14}$  whose pairwise squared chordal distances  $D(a, b) = 2 - \text{tr}(P_a P_b)$  are all at least*

$$D_{\min} = 1.728603 \dots$$

The exact value of  $D_{\min}$ , a rational number, is given in Appendix A.

## 2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some  $N$  in the row  $m = 14$ ,  $n = 2$ . The cell  $N = 63$  is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 1.727385, obtained by deleting 1 of the 64 planes of the packing found by this search. The Rankin simplex bound for this cell is  $2(14 - 2)/14 \cdot 63/62 = 1.741935$ , and the value above lies below it.

## 3 Verification

Each subspace is given by an integer  $m \times n$  basis  $Y$  (coordinates times  $10^{12}$ , rounded). With  $G = Y^\top Y$  the projection is  $P = YG^{-1}Y^\top$ , so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m14n2N63.txt`

in <https://github.com/Ge-limin/math719>.

---

\*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

## 4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 135 seconds on one laptop, with about ten such searches running at once.

## A Exact value

numerator:

108037747365972206694626881380532776197712026904677928214913613170620362  
404404667967679227071942

denominator:

624999999998055353299574819547502326881635465275875645690514540996330409  
44270248743205342516137

## B Coordinates

63 blocks of 14 rows with 2 integers each; divide by  $10^{12}$ . File: `grassmann/packings/m14n2N63.txt`.

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| -115791181856 | -346925075391 | -143821673082 | -75622975949  | 66094018365   | -153413199804 | -340066836270 | -3340178735   |
| -172870318951 | -373143296459 | -328127527020 | -62098748557  | 258546917166  | 38104479487   | 308511025808  | 7618572205    |
| 623944010180  | -414640920693 | 57981922172   | 54130760554   | 290622327744  | -144027957942 | -169779026848 | 273263063457  |
| 156664127523  | -236277165679 | 275881380066  | 150187987789  | 488085571944  | 22376504752   | -79850866472  | 125855133017  |
| -98918624535  | 124621449918  | 290887485431  | -450075395590 | -206845137494 | 417887922785  | -193228925897 | 399579274288  |
| -378682436691 | 137154756784  | 42515734130   | -238963243253 | 37249307447   | 360117829735  | 372252924472  | -188183851487 |
| -273166386430 | -212842371285 | 387266338921  | 607364465771  | -305109737891 | 182148457522  | -178525774317 | -447588632318 |
| 47344565038   | -309611774875 | -357408257949 | -100872422083 | -237852509107 | -244945226365 | -634154891187 | -165865816435 |
| 124193086112  | -143580823440 | 298934003528  | 134177030089  | -301164427300 | -236758411587 | -200433079100 | -115740047705 |
| 93607379383   | -38664983004  | 351778294748  | 106636262317  | -334647345977 | 334389183437  | -26449269576  | -34419347843  |
| 84647931534   | -228544985841 | 104348661085  | -63111629756  | -21846438212  | -245006717227 | -121107777103 | 224434339315  |
| -103047181931 | 64581012712   | 138597053776  | -379011929388 | 354783060808  | -297633996523 | -17204846612  | 109356911585  |
| -115797819330 | -210331955206 | -218861010891 | 66989993989   | -84500430549  | -332667696261 | 222192441578  | 76898200871   |
| -507377985918 | -463029521832 | 300117100201  | -76747180015  | -112042733662 | -162784012317 | 99911518908   | 176004701553  |
| -111356546723 | 64732929544   | -376634165570 | 336646955294  | -213760291326 | 81158615601   | -102930862874 | 382632291701  |
| 569928746706  | -306224918577 | -167898709140 | 48912734377   | -179008896455 | -250401065326 | 592371535639  | 127854397865  |
| 248797330406  | -171740580067 | -92355219444  | 188669366520  | 272692950252  | -110590971228 | 380463763706  | -320905082844 |
| 308299394872  | -185838294486 | 51579281801   | -91096242798  | 60708773742   | -548121738524 | 217838905090  | 282819642248  |
| 221271694973  | -360203486832 | 330322154745  | -225427908613 | 582571498323  | 236363756689  | -340882925971 | -278086070135 |
| 386455784945  | 265395034801  | -198287653892 | 150855842791  | -80180989918  | 123892464797  | -179162855168 | 73773996082   |
| -8925046220   | 117963164721  | -462677059080 | -82562864054  | 98002821052   | -424309704264 | -267343526479 | 36932172558   |
| -228238740268 | -87774234537  | -143730662672 | -140112974038 | -433039505397 | -340865703357 | 128637251643  | -679352198588 |
| -288304724597 | -71122617756  | 157844508803  | 261579846099  | 353641828622  | -200825619252 | 367368586885  | 90639469953   |
| 305210676801  | 195708569916  | 269320919478  | -186853529128 | -17942016914  | -206873921260 | 433630831658  | 218635008351  |
| -13932595486  | -366917689917 | -59860657389  | 270395129872  | 119751647954  | 131783130146  | 101525151340  | -182664668563 |
| 218051147336  | 417916356520  | 371629826321  | 622620769073  | -55553188986  | -632511024364 | 412760359302  | -242879748106 |
| -116170150416 | -236163758078 | -166048165433 | 95845318493   | 183552195388  | 252721848343  | 79428349122   | 323446318935  |
| 99772725523   | 455838001584  | -124104313610 | 420655420723  | 52647158474   | -289356237191 | 529132838763  | 48709319307   |
| 320302238727  | -362804101379 | 537153678054  | -265784228787 | 568044115953  | -133995438754 | 2127770430763 | 191859185472  |
| -349057640068 | -99031352474  | -192383320381 | -249575221427 | 36382959947   | 112196607720  | 95075501644   | 182287913377  |
| 123035717945  | 222660771610  | 80748568076   | 92212590660   | -434345023479 | 117138866421  | 124128094448  | -102846059842 |
| -108402271429 | 422467105052  | -370139617276 | -245258062774 | -168937447442 | 72000477851   | 294640967911  | 289436561337  |
| -396020610207 | -161239275211 | -313920433852 | 397774337635  | -283465363170 | -58725045916  | -29856383742  | 125335967962  |
| -91204982789  | -485261455243 | 78564073526   | -22241425765  | 45300320731   | -52102536282  | 343211699887  | -157188227069 |
| 21444823937   | -88490032667  | -111975414162 | 568308019589  | 276225675022  | -176144317513 | -5911647253   | 531008476231  |
| -35193026696  | -123457120398 | -388600030920 | -181426245259 | -293624591889 | 49398048471   | -240078139094 | 172387674685  |
| -67142097243  | 190002451603  | 119181778042  | 12630164824   | 241089918261  | -336761305580 | 149364894434  | -486165856101 |
| 225723313591  | 456001780753  | 17496008423   | 298813842559  | -559618811    | 395242902157  | 153908098831  | -223112302804 |
| 38315465593   | 98154735188   | -248049877261 | 391288437989  | 296207530811  | 277298863874  | 103040802000  | 163257767007  |
| 662239037315  | -108362410080 | -615859025792 | 168849536547  | 491980257348  | 17624992208   | -75165735237  | 54232820582   |
| 293628052430  | -201356359302 | -149757499049 | 88711463522   | 294840320881  | 214613440408  | -158090196725 | -94271541648  |
| -11053103508  | 265089305172  | 294247627810  | 195346282277  | 118715987269  | -333657495317 | 174345654514  | 5539261957    |
| -204039641730 | -213246275285 | -75676493544  | 122838370470  | -109149122582 | -249553481779 | -790356256861 | 135985666888  |
| -225893779750 | -648475064006 | -136485473527 | -118188271231 | 391689584376  | -282413050003 | -122608388429 | -103119235147 |
| 121069859491  | -224045638045 | -77744163367  | 281222727753  | -401655474359 | -391767237525 | -158801448360 | 336868975438  |
| 59448705881   | -95426010413  | 132349978288  | -244585445071 | 72676402130   | 61963381152   | 143379992475  | -38487399726  |
| -79045426785  | 35140342794   | -174563555588 | 318601825978  | 100117523556  | -385079197370 | 241659694743  | 591280155375  |
| -166166896147 | 320967032088  | -196404036841 | 90367536726   | -81146649686  | 121141413245  | -193527619084 | 294816320208  |
| -345441353793 | 39369128840   | 488738988667  | 168788298582  | 397231918986  | 600487378000  | -29457098733  | -227048011895 |
| 452725089751  | 71526053767   | 272938498610  | -58046883629  | 299295176631  | -89972200294  | 181994645677  | 195891920165  |
| -355113161125 | 449055667336  | 70350613377   | -581289276405 | -230303324886 | 316306256931  | -58635435371  | 491088385461  |
| -270880566433 | 274205987894  | 321065603607  | 18091113246   | 346987129453  | -146345097479 | -123672015996 | 243464779249  |
| 218891195992  | -15703293376  | 45407351494   | -309641905616 | -163126185375 | -235643460016 | -377574647407 | -116912013302 |
| 467714877010  | 177041220484  | 107177129708  | 27317471731   | 168854483877  | -170959000104 | 61471740673   | 394776275865  |
| 101687257757  | 230441662294  | 324431160396  | 252206983132  | 283323519359  | -103503331874 | 96795321414   | -269261904584 |
| -233465016585 | -47428882966  | 242714754002  | 75957410486   | 160250751630  | -449787137438 | -121368630932 | -109083753010 |
| -242154801938 | -383678504125 | -239729814984 | 105534322724  | -51263609354  | 42215489538   | -275066404695 | -247572020749 |
| -124916018000 | -87594423032  | 456739338838  | -288662414721 | 234047874840  | -145911740040 | 314307079507  | -565603840089 |
| 288346784183  | -106806972806 | -219651876268 | -409333401010 | -587944945582 | -94355462287  | 263446780684  | 257179707983  |
| 25879788173   | -149763283364 | -385511585571 | -446551603101 | 9299532171    | 320626874970  | 72642108273   | 103210406584  |
| -161591124474 | 136699008197  | 2255969054    | 299119685079  | 78545191302   | -276234761313 | -339007015525 | 266944467216  |
| -222347769722 | -495265393963 | -99613522643  | 330355687840  | 97185216920   | 31536877072   | 157758745726  | -21166296433  |
| -369157731661 | 26981216302   | -472931437281 | 275586016456  | 155178103406  | -384097937891 | -541758538243 | -57436257780  |
| -152428647503 | -366203877798 | -207715122739 | -362218081133 | -109051679937 | -290606692747 | -274174993123 | -365948424503 |
| -325014760198 | 332211038273  | -137565092693 | 114270039905  | 122777869438  | 87775160480   | 238001672185  | 141028344092  |
| -179867037455 | 45949409205   | -292229369438 | -219523429713 | 195520877809  | -219523429713 | -269355760238 | 65184248530   |
| -520468452701 | -127802930421 | 179568332549  | -69084015982  | -120797883023 | -430372540445 | -21972524211  | -398158082164 |

185276832755 73302193054  
-186955721802 79014556717  
337291109869 -36261850302  
-14574504011 45922709059  
249922302737 -286167357347  
-28521083879 30887324285  
517414299477 -61144596347  
-562199238772 -182412200937  
178105914228 -482597220556  
69159013265 -76136067311  
143538600903 366644757393  
-198843808591 -347010646951  
-93766984159 -757756334279  
-251815126169 419789885145  
4317900543 -71874493858  
-134374668019 -159846338857  
237244645291 171808462521  
-157901096921 151371610645  
-33540077001 -49405465261  
43216191091 -259034638620  
549378679318 -57842335775  
-14878789256 -216607899988  
75660435131 -159077276243  
-111160312832 84983239276  
381247835126 108869296335  
598236148282 47420150759  
279902370038 -40978291727  
6573331171 -229367243993  
-245074178047 109802403171  
-478046888262 -13629789261  
645375661885 -308623781797  
140541213864 -195653580801  
-51661355620 -122138554344  
-193509490944 -213818224866  
262176137112 533784440100  
-19893390259 -588359279552  
-1755191930 -391830855871  
-176958901275 -154247866477  
-32982435081 -268316334101  
234749715517 104356322272  
171038253735 -237001453717  
220590901224 517866107320  
-10592463913 -357706556797  
-231996531770 -110689741075  
-572612383964 74191109775  
-16178626722 281137687434  
-250784534084 -77757268089  
106060441531 -111134953828  
-29428562010 452001201496  
-238391199556 205528884699  
-273871574915 335380261139  
-164913872016 -206749358899  
469586838617 50104832618  
259777838749 167976175942  
-27384592955 209953468374  
-31736627365 -188207646442  
-361649586422 170392151366  
-1395547181 291481013825  
-78750715228 -506284961370  
-217457445150 -59233483986  
-237003061847 115366263947  
323181681635 112479698317  
295883610195 366592402190  
-354294805397 301992206300  
-502000601242 -303547333094  
400541376450 -248428608958  
-126622756991 374744883451  
-107873908327 28772630873  
-61994108089 -146022447939  
59111475942 -242187572261  
-113118348615 429850002021  
-161235816766 -183984090497  
86000715454 282504210338  
174391532561 -324102723699  
7478762979893 -116540775230  
-323112976979 -263297780435  
-233688109315 -34312993033  
-33647150551 -40849267582  
40154452000 -84262112499  
-187707124252 -491243349733  
-371867260161 117814217553  
-141913431602 57388340991  
447942408830 406045549377  
-61004977505 330201081410  
-495125456204 53265709692  
24256501824 -351457352820  
254070896586 -395511091074  
-167091230171 -418987049401  
-311100541938 -7227707675  
65458738194 41015573106  
-133178730572 379129684828  
113634051731 164631150907  
371643391585 -54126801409  
306510554693 59844287759  
249391090563 -77493269236  
-167543258069 287510127319  
-2014774422 -255350015214  
-428572158461 359063541663  
-57936757860 -232465744085  
145102673215 -94077662507  
283899163858 86185313214  
62808673783 -101082492564  
-431051804619 -245135654332  
-279225374459 -614685460648  
-230500987416 -199166634133  
-519812893425 404825409484  
236435205914 -155522216037  
-99042991605 -63580657902  
-63735257951 -2220859313  
224688241756 235834642658  
-272123217412 -448173789103  
208691850784 -148405987844  
286957381315 -306210238424  
-339403772111 -14340369489  
46090561846 -26099272401  
-417936386830 25179482770  
223476318121 238685555402  
243638487324 159505926838  
-510122393313 -78585731425  
149384524463 -472684327950  
-270283491417 186208811455  
184100067608 -175987404183  
-40130953128 -368601704279  
87541874667 -384764670054  
221400557554 202005131816  
110802508536 155255799841  
104904304708 249076750106  
459146670753 -328074272962  
-391000987867 -98650767410  
-283737265686 80213570814  
-272485166580 276028150595  
-387459392619 35623344571  
-130677009242 31294877793  
77076607354 -14532101469  
101976993080 -616633695684  
-98946220823 -386358593169  
164684037271 1122168809121  
-434505104008 -356002765059  
401757131651 -250193172749  
-176275799616 316885037460  
-68168099258 173228252061  
-160338746907 -540756843838  
222678189062 -242264605921  
410727531893 457253336623  
-365399094947 255136260196  
-255165879935 -323314881782  
188408262031 35254195935  
-50452347651 78818737678  
-456614700359 -74989676960  
250871373765 -109588076412  
-218247178082 212835995477  
-56078794838 82359309755  
447041553909 273545732924  
-243191369773 61082860918  
104044888755 137274436371  
-306489815675 -108861327836  
-88711574917 613795563985  
121101613719 -541548267306  
675129468999 -2249461432  
-12536658640 -102229793845  
188220622096 -137240916056  
-147256515132 105386740464  
-24262391132 -52276232622  
56538143894 266577407486  
213738628573 -257765298631  
226223867336 200225730520  
159562692776 25134553548  
261740530794 409638204826  
235123917003 159276559480  
-20492481309 13527163977  
-383165631675 -467679491675  
191490505680 165075353414  
112948303404 253228703503  
-184498309353 170272868255  
230966529738 -2295793734856  
53700608096 -232887828168  
-130091442290 -38927887550  
258795715606 247170597394  
646605950964 -440472460496  
-248635130775 291954806265  
-192237285604 -11738739051  
-556256484038 245664247550  
196327524539 334788581257  
361511071355 -119952499913  
147980871465 411507559853  
-94245157848 -213105716807  
341891860409 -57058524149  
-173734047908 371404756351  
156555609256 32903064211  
-33029529738 251539394735  
444062999405 420648642311  
-147580084549 359213242939  
23951303056 280478687280  
247767318938 -38500905973  
-116792602245 -429906675060  
-429301856949 -491006461248  
433373312660 154863969395  
-2412181618 -142328784938  
-7057393381 -401656566320  
104869407302 103862889853  
-141207204526 11631338385  
-429724882436 167475007325  
14573063953 32094122179  
379474186809 -367181284628  
-450551413508 170713501090  
1464618650 -116292526137  
202341586753 161352827050  
-102161326596 414133709936  
-42643903456 77179528430  
131455981985 -357252260522  
-388113102967 -231060693632  
-314115301663 -123642842203  
-509891798467 137824436554  
359348991542 -188815735010  
-243947116180 -212878740166  
-205711959626 -206601980572  
-84283507166 340683095613  
-109713426569 -348068014156  
-441160895921 173850488580  
346300115647 540156823580  
-297920687215 298747870505  
119877153622 81417823592  
296689877584 563415418616  
348770341594 140337892711  
350413249428 -85745870374  
22594793991 84463801361  
201486463093 212054842430  
303172083606 -240963732364  
421084460208 -256817453728  
71846407476 363294110967  
-85907268179 -385136502554  
-29381402459 -88838639862  
55191531189 -218087933761  
-11563643981 -28363363134  
263321733786 223948952640  
-520853890542 304418908172  
111954029613 326556637503  
92093390388 174803329386  
-432900155087 429837369767  
179642901210 -260167396837  
60632443511 28102300837  
-383777578382 -208246268828  
149024948890 2627595450  
299034743938 311400723103  
-316858622176 307383989497  
-179443686917 -579050368367  
-242373124031 -156780193415  
-372462368957 -6515958818  
-275900757701 22150582459  
-302037174681 119692840798  
126742188530 226401612527  
-390474381542 366696134982  
205876434806 587221266640  
73325776227 -355464434164  
245268176643 304014129684  
261277531229 -129886883810  
-94114792215 -6417848778  
300840717505 -361376065740  
-185612859107 -164802592955  
-96817576413 79095849903  
-278239806285 80263022504  
-433790325157 27032761335  
-435050826672 -99736055629  
-239233790587 -229406144866  
1666489889933 -382657829838  
-364587542312 173306198829  
-370610479276 -401490677236  
-128648575652 160425404844  
-337778528808 378150710608  
-134903195913 -308542530404  
-434421893602 120603378768  
360197843410 120339517022  
352051723690 101041080000  
380329624026 456014021267  
-64644103075 -73446176174  
-120379609611 371343966890  
-73993478400 84176550132  
-214444242008 29140751234  
252114628878 -55307888724  
-363153966106 -37214124777  
-259199827321 -581106041871  
163116941195 268955653988  
176517484406 508206415896  
-473669081917 169034702214  
187149950104 90572879038  
-381651787542 251071666901  
-203436708947 372111343510  
74923636601 -221135204782  
-896772900922 -14060077774  
-178713773366 181781517041  
38131473953 82041542517  
431425328963 -10353715071  
-110726932269 140060655234  
438578249462 -211713754728  
-437869533047 198737215254  
-436636742852 -427897160671  
308481475739 -130484315198  
-338963163069 191594207467  
11486612481 138177229306  
111154922090 -198657362711  
-163104146574 -70112467899  
11178285474 369015600129  
-214356716030 -119671404572  
-204650338283 31327647157  
-171272865715 -663304296302  
215801326998 -65666730067  
179192414753 -129232100340  
182818449611 -103244524456  
-98649537735 230382608015  
104883683010 -110791369285  
324314522122 54390788401  
-353328893294 -523775171396  
-220770460880 -359098578429  
-442106221395 -99425096394  
277856416319 591191264655  
287218985 271555020582  
512805840133 -175770527304  
-181846653803 480930989179  
169228593834 -225139163869  
193999557991 -163164481928  
-54564781832 -415744914378  
-123041389317 93504504263  
117064064506 242748710689  
-337581391007 -152552210512  
128166381394 -13513451519  
-166968029016 -10096881716  
255354347947 -97344246140  
-172112705339 98990934919  
138456553916 -374877201302  
767671970766 -101714236849  
253939312447 -131076715292  
-8495704460 -456008762953  
43469020402 33587256299  
-200374069665 -477326004700  
509341491678 -46042828562  
161790466351 -253767557046  
-402154793313 102027254744  
199883090508 258854771506  
-203825522808 103264975512  
23677894563 -45614336714  
-38494858368 -240011271438  
72229783626 597500875694  
151008224692 6946557527  
-362155380024 486351417357  
-21184929071 -87021412417  
-164875381729 -3846163639  
328773513824 55820699482  
-417399122058 -425934591541  
115525754290 331313522191  
-198654376106 -186639477592  
21062663577 79652781028  
190995283081 -65815126749  
-266962472008 -112127106213  
313341304967 -200597188694  
64608233798 -323921572577  
-379560447406 97587434214  
86230865666 -477463074909  
-324968801937 7394461116  
-183847830361 575030004377  
303247479530 -76995023896  
398432501797 202838756995  
441618966465 267775427350  
-552191531678 173220864590  
-149877972311 -30700770431  
25177637429 -393078470492  
510165482984 193546423328  
215565354840 -210499041783  
-234099291804 -136494257486  
-163353877823 316426790628  
-206653884263 309041884586  
312445596953 461714653595  
-99811276609 -54128593000  
74062932982 -16714696721  
245765193733 -3333537121  
35728035678 -547108293457  
-256930640791 -40979862094  
200269268924 -244341607006  
-115033599480 -245886562493  
-37734689151 722767116856  
-188830204052 -246122093873  
219113445421 -88115454544  
-26625400645 -64004131665  
290245813949 470912368424  
332205794969 71381730160  
217479072715 -81142415883  
180747613887 -55558522575  
152723360857 110168062794  
-746848047406 166782114388  
24543510338 11485170830  
67671651623 -91639908472  
352474649314 11291915305  
-309284400580 -41822436986  
-416608032433 155462662274  
-63386275342 -323378630559  
-280388646436 352923155645  
-111048383684 623486865320  
435496528321 41371306846  
-60459768778 -152560946312  
-320767166802 141025950498  
-277864790195 15471548864  
359198019257 312912930381  
94774738419 -123636500954  
-169868953595 -137158072887  
235821438321 -76164551673  
-41988586438 -28266792569  
-176047533469 -38968318442  
-98687698210 -24108842201  
-309025298942 -688026569944

|               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| -12483696806  | -116474662316 | 618486442147  | 123582762434  | -347845891567 | -486677122299 | -257025581253 | 463004332431  |
| -112354783692 | -235379807166 | 46106054119   | 355826836740  | 30618223639   | -298729317302 | -176051206212 | -166385626056 |
| -81527712585  | 198778381169  | -192661271358 | 125167863770  | -187004296216 | 216972682763  | 7290908362    | -430162417865 |
| -583508419948 | 188626525977  | -92998957719  | 371157563240  | 488198693005  | -296735486547 | -264984868050 | 244762421143  |
| 79157588966   | -33077189027  | 45334852194   | -157099441589 | 199980431869  | -82672242919  | 159569889942  | 142953325137  |
| 211163566432  | -52980438342  | -45529463312  | 164876844049  | -125993486177 | -431262462663 | -13393871071  | 53471123166   |
| 315469959445  | 184269305621  | -297259367463 | -226614628462 | 439112798973  | -40337859717  | 197700535938  | 437830185285  |
| -562033277624 | 191029948640  | -188453666274 | 103446282542  | 77546649816   | 116855710805  | -444561233400 | 161377586632  |
| -189259623687 | 314351846317  | -226555373507 | 250328200320  | -767247356676 | -469497774790 | 402959494855  | -95879783257  |
| -4707758436   | 454414041193  | -208041801206 | 451402632958  | 52305405655   | 18606601171   | 32130483575   | -172542657764 |
| 368629982348  | -125684860823 | 91752028169   | 546139130770  | 169008935453  | 261111801084  | 47464795813   | 292445757288  |
| -94908231151  | -55960839382  | -245183092094 | -73008454977  | 195137934793  | 236738189468  | -49708555531  | 127068666865  |
| -319481870371 | -303569935095 | -502931342744 | -70098890401  | -292242252657 | -174513243755 | -256114728249 | -115963238750 |
| -116037106204 | 215330771124  | 159667270146  | -125813817802 | -1482416333   | 314822484676  | -246852178736 | -626691221400 |
| 120980637362  | -97857196668  | 98527272787   | -114387454528 | -348984547714 | 559474881819  | 299140079306  | 182922507878  |
| -34584461924  | -138948122344 | -75726735303  | -174523271486 | -183514984030 | 48960249452   | -244738728560 | -112577982168 |
| 12820759773   | 117552714049  | 37973712669   | 148322907021  | -171300956650 | -40982697425  | 564492287286  | -177448014022 |
| -379386740370 | -737849903138 | -554869922726 | -341401835246 | 632309139073  | -15491265678  | 100693411851  | -367349175946 |
| -432602200113 | 422397333123  | -521044520438 | 385995885736  | -240246038588 | 158153846928  | 264538354580  | 99814865310   |
| 173862474532  | -42036818925  | -343754403672 | 86747516347   | 417959771338  | -327052717966 | -9992234697   | 150138432068  |
| 74170591086   | 30952237719   | 41116816407   | 381368070040  | -380577721978 | -488724227419 | 15063141247   | 334514020261  |
| 495061541680  | -91296358146  | 85058522775   | -617587725852 | -262971305736 | -379602414720 | -396051582629 | 310167575772  |
| -333885842587 | 222409035891  | 49592385235   | 133696168175  | -348987804799 | -45553042760  | -157923105998 | -213737532515 |
| -14150331578  | -128376304257 | 38882403921   | -35652658170  | 42446021436   | 260192562459  | 284405335778  | 160744908433  |
| -36794245787  | -47388210315  | -157250820139 | -27634612271  | 239481680310  | -307245165134 | 201600129251  | 249164641126  |
| -43587095490  | 1878089361    | 158121351460  | 246634195423  | 372598962703  | 271357760424  | 157451646008  | 378183890403  |
| 378947660764  | -319800924567 | -88194511170  | 214339853751  | 21448047857   | -150339693610 | 33443252615   | 183218569083  |
| -10116186519  | -242847466379 | -463870107720 | -88592506801  | -241205252992 | 405675377512  | -403363155128 | 156625306896  |
| 474648877517  | 365089645502  | 161244102993  | 16815690624   | -69936630591  | 158039513085  | -295649860325 | 56091745561   |
| 269819745095  | -207786736470 | 277303654331  | -233698607897 | -39693787868  | -19224581482  | -81400957377  | 511093449871  |
| -418376685439 | 519498423131  | 118680288097  | -170151481993 | 243974399037  | 27413141342   | 305635005050  | -909114423    |
| 131391389405  | 102742313183  | -352681362395 | 58526584864   | -330442606553 | 184370696694  | -293227003769 | -114044467105 |
| -256805615573 | -92892555370  | 376957886646  | -51172295171  | 283383898401  | -43890045064  | -105775469201 | 493893449675  |
| -66762772999  | 78582198598   | 211723342551  | 304579455307  | -205768680276 | -32868608450  | 387308852007  | -7654128195   |
| -218138901723 | -241955723576 | -177468103725 | -560549803586 | -203433112726 | 99895414312   | -52271048876  | -349612964580 |
| 422367505664  | 24224270906   | 169743481112  | 265951538938  | 149933365781  | 363379277563  | 477033889222  | -147249522045 |
| -40014938697  | 207041732850  | 94038236889   | -53766656789  | 440347190391  | -247782412724 | -251778215706 | 30632884450   |
| -254281741780 | -512643228738 | -267580974601 | -35245636438  | 36248828940   | 352942891309  | 57264387947   | 391198778979  |
| 85223684247   | 479353029570  | 403862837008  | 395737183579  | 363595939557  | -95306575330  | 139748372432  | 208084634602  |
| -29656348205  | 288442476973  | -196398552560 | 129221037181  | 555456703853  | -15053289965  | 702592680     | 195766229396  |
| 451646393353  | -170649170614 | 475510901631  | -446820071994 | -143582275166 | 333304534814  | -5661070132   | -218870437996 |
| 414949896520  | -122390872867 | 56312529492   | -235169031931 | 199438794902  | 590805542442  | 532989320125  | -23321024954  |
| 70948534700   | -195579761577 | -137780573059 | -297555294568 | 234994908176  | -195207959625 | -169134795482 | -90723193587  |
| 137007062920  | -314910377655 | 124493769529  | -387929633125 | 11032908247   | -110654225900 | -368792445624 | -111707995812 |
| 526638165733  | 319654040489  | -166517267517 | -141646910396 | -147767368521 | -334885303327 | 86119460518   | 242446439932  |
| -18996511618  | -300911024133 | 106647342001  | 460066005742  | 199906873891  | 192492073697  | -251912412541 | -286108614372 |
| -16776729163  | -406463797567 | 396549582671  | 78061146741   | 573581419143  | 7128615687    | 234806174727  | 492684759096  |
| 248424292384  | 223573694492  | 100816558492  | -151508893664 | -193569849209 | -489212374505 | -515525530462 | 445351228077  |
| -6733704477   | 188552024616  | -414194475732 | 176617915160  | 370902158013  | 12233890961   | 161806691366  | 28265165893   |
| 428901230159  | -81884202279  | -70094549974  | -55765888128  | 130201139442  | -107947280786 | 217062124680  | -335753888611 |
| 60898426374   | 102837077559  | 305932836643  | 65074546846   | 221319240544  | 143548931735  |               |               |
| 253909382351  | -211528810082 | 306092450680  | -20905567566  | -18538159213  | 13172258802   |               |               |